OPTIMISTIC RESULTS ON OPTIMISM DETECTION: LINGUISTIC PECULIARITIES AND STATE OF THE ART PERFORMANCE USING DEEP LEARNING

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ABSTRACT

Being optimistic saves one from current painful perceptions, while being pessimistic saves one from those feelings in the future. Although sentiment analysis is a popular topic in the computational linguistics field, not all types of feelings are enjoying the same attention.

The main focus of this work is the exploration of optimism and pessimism in social media. We begin from a dataset based on Twitter posts. Our approach is based on previous studies targeting the same issue. We both replicate previous work and extend their reasoning. In the first part, we introduce the main datasets used in our work.

In the second part, we explore linguistic peculiarities of optimism and pessimism. Thus, we observe correlations with verb tenses, pronouns, special characters, etc. By doing so, we build an intuition on the differences that characterize both optimism and pessimism.

Finally, in the third part we apply deep learning models for optimism and pessimism detection. We analyze the importance of pretrained language models when the used datasets are small. Most notably, we outperform the previous reported best models in optimism and pessimism detection. We also analyze the feasibility of transferring knowledge from larger datasets, though labeled with other feelings. Thus, we further increase the accuracy of our models.

ABSTRACT - ROMANIAN LANGUAGE

Optimismul poate oferi detașarea de neplăcerile cotidiene. Totuși, pesimismul poate evita prilejul neplăcerilor viitoare. Deși analiza computațională a sentimentelor poate părea un drum bătătorit în contextul procesării limbajului natural, analiza optimismului este încă o cale lăturalnică, neexplorată în profunzime.

În această lucrare ne aplecăm atenția asupra detectării optimismului, respectiv pesimismului în texte postate pe rețele de socializare. În particular, ne vom folosi de un set de date format din postări de pe rețeaua Twitter. Abordarea noastră se bazează și extinde studii precedente cu privire la detecția optimismului. În prima parte a lucrării, vom prezenta studiile precedente și seturile de date pe care le vom folosi pe parcursul studiului.

În cea dea doua parte a lucrării, vom explora particularități lingvistice ale optimismului și ale pesimismului. Astfel, vom analiza corelația acestora cu timpurile verbale, pronumele personale, caracterele speciale, și alte trăsături lingvistice. Prin această analiză, dezvoltăm o intuiție cu privire la diferențele dintre optimism și pesimism.

În cele din urmă, vom aplică modele deep learning pentru a distinge între optimism și pesimism. Vom observa importanța modelelor pre-antrenate, pentru performanța finală, în situațiile în care seturile de date disponibile sunt de mici dimensiuni. Rezultatul principal al studiului nostru este depășirea performanței de referință pentru un sest de date bazat pe optimism, folosind modele deep learning, pre-antrenate, bazate pe mecanismul de atenție. Mai mult, vom analiza posibilitatea transferului de cunoștințe de la seturi de date, mai cuprinzătoare, care nu sunt adnotate pentru optimism, ci pentru alte sentimente. Astfel, vom îmbunătăți și mai mult performanța modelelor noastre.

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ACRONYMS

ALBERT A Little BERT

BERT Bidirectional Encoder Representations from Transformers

BiLSTM Bidirectional Long-Short Term Memory

CNN Convolutional Neural Network

EmoNet The Emotions Dataset

FastText Enriching Word Vectors with Subword Information

GloVe Global Vectors for Word Representation

GRUStack Two Stacked Layers of Gated Recurrent Units

Hate The Hate Speech Dataset

OPT The Optimism Dataset

RoBERTa Robustly Optimized BERT Pretraining Approach

Transformer Attention is All You Need

TSA The Sentiment Polarity Dataset

XLNet Generalized Autoregressive Pretraining for Language Understanding

Part I

OPTIMISTIC INTRODUCTION: DATASETS

Learning mostly implies acquiring experience. In the context of machine learning, in many instances, experience can be referred to as data. These days, at the bottom of the machine learning models' performance we can find valuable datasets.

In this chapter we will present the work performed until know in optimism detection. We will also discuss the the datasets used in this project. In particular, optimism and pessimism, emotions, hate speech and sentiment datasets.

1 PRELIMINARIES

Optimism and pessimism are feelings that have been investigated in various psychological studies. It was found that besides contributing to the well being of the individual, being optimistic also has beneficial influence on the close acquaintances (Peterson and Bossio, 2001; Scheier, Carver, and Bridges, 2001). As well, optimistic persons are proven to better manage the stressful situations (Carver, Scheier, and Segerstrom, 2010; Scheier and Carver, 1992) while being less anxious or inclined to depression Achat et al., 2000; Rasmussen, Scheier, and Greenhouse, 2009.

With all these pessimism and optimism are not mandatory two diametrically opposed concepts. One of the most renowned figures that intertwined optimism and pessimism was Saint Augustine of Hippo. In his influential work, *City of God*, he famously exposed pessimistic views, nevertheless in an optimistic text structure (Lamb, 2018).

1.1 RELATED WORK

Computational analysis of the difference between optimism and pessimism did not have a top place in the research community. Thus, even though sentiment analysis is a well established topic in the natural language processing community (Go, Bhayani, and Huang, 2009; Ortigosa, Martín, and Carro, 2014; Waterloo et al., 2018), there are still feeling such as optimism which are computationally unexplored.

Nevertheless, the task of optimism detection was first introduced by Ruan, Wilson, and Mihalcea, 2016. They built a dataset of Twitter posts labeled as optimistic or pessimistic. Starting from this dataset, Caragea, Dinu, and Dumitru, 2018 explored the feasibility of optimism detection using deep learning.

In this thesis we seek to continue the work for tackling the problem of discrimination between optimism and pessimism. We will perform both a linguistic analysis of the main features of these feelings and we will extend the field of deep learning models used for this classification.

In the remaining part of this chapter we will present the datasets we have used in our study.

1.2 OPTIMISM DATASET

The main focus was fell on the Optimism dataset (OPT) introduced by Ruan, Wilson, and Mihalcea, 2016. This is the only dataset, to our

knowledge that was annotated specifically for the problem of discrimination between optimism and pessimism[1]. It is especially of interest since it annotates tweets. Social media becoming an increasing hot spot in recent years.

For each tweet annotation procedure was performed by five annotators. Each one of them labeled the tweet with an integer score between -3 (very pessimistic) and $+3$ (very optimistic). Thus, for each tweet a golden standard was established by averaging the five ratings.

In the work of Caragea, Dinu, and Dumitru, 2018 there are two definition of optimism considered. The first one (Figure 1a, referred as the 0 setting), considers as *pessimistic* the tweets with the gold standard less than or equal to 0. Same, a tweet is considered to be *optimistic* if it has a golden standard grater than 0.

The second definition of optimism (Figure 1b, referred as the $1/-1$ setting), refers only to tweets with a more clear-cut annotation. Thus, only tweets with scores less than or equal to -1 are here considered *pessimistic*. Whilst, only tweets with scores larger than or equal to 1 are considered as *optimistic*. The tweets with scores within $(-1, 1)$ are thereby ignored.

# Tweets, setting 0		# Tweets, setting $1/-1$	
Pessimist	**Optimist**	**Pessimist**	**Optimist**
$2,796$	$4,679$	$1,340$	$2,507$
Total		**Total**	
$7,475$		$3,847$	

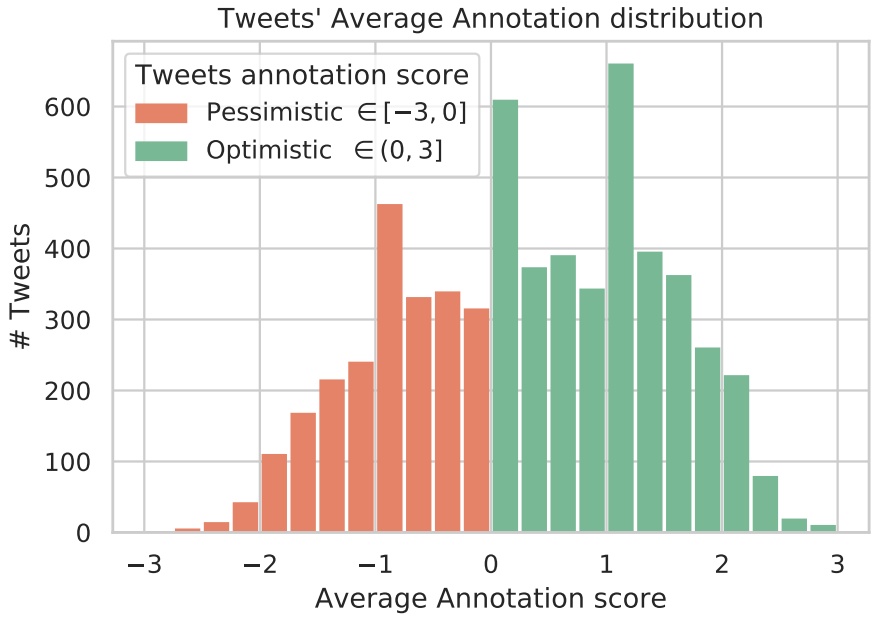
Table 1: Number of tweets in the OPT dataset, by Ruan, Wilson, and Mihalcea, 2016. In the 0 setting there are almost twice more tweets than in the $1/-1$ setting. In both settings, pessimistic tweets are less than half of the optimistic ones.

When considering the 0 setting, the OPT dataset has a total of $7,475$ tweets, as presented in Table 1.There are $2,796$ (37.40%) pessimistic tweets $4,679$ (62.59%) optimistic tweets.

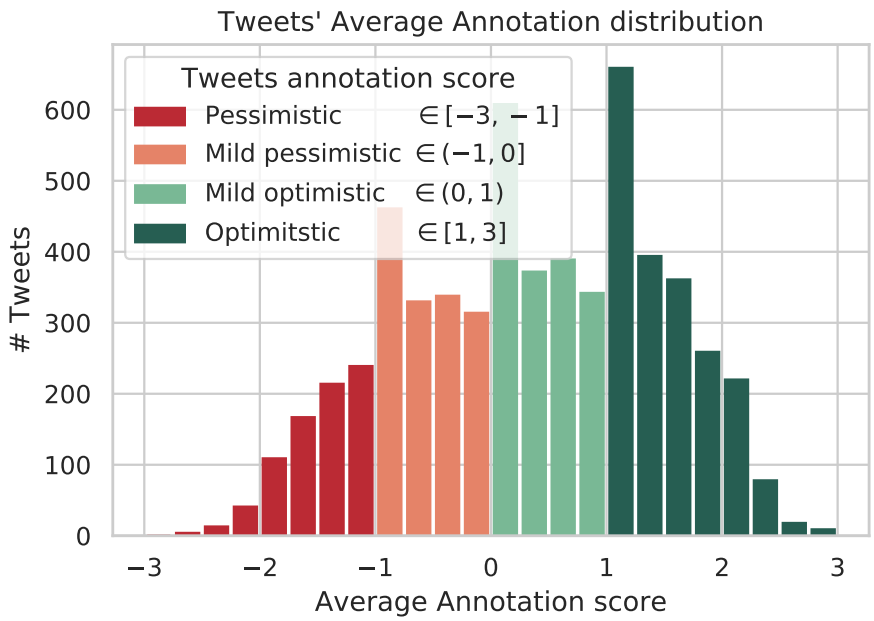
On the opposite side, when considering the $1/-1$ setting, the OPT dataset remains with a total of only $3,847$ tweets. Out of these, contains $1,340$ (34.83%) are pessimistic and $2,507$ (65.16%) are optimistic tweets.

As presented the OPT dataset is a relatively small dataset for the straightforward usage of deep learning techniques. Nevertheless, we will see in Chapter 3 that the usage of pre-trained models overcomes the dimension of the dataset.

1 The OPT dataset is freely made available by the *Language and Information Technologies* from the *University of Michigan*. The public link, where more other interesting datasets can be found is: `https://lit.eecs.umich.edu/downloads.html`



(a) 0 setting



(b) $1/-1$ setting

Figure 1: Optimism annotation distribution in the OPT dataset in both definition settings. In the 0 setting (Figure 1a) all tweets of the OPT dataset are considered: pessimistic ones have annotation scores between -3 and 0, whilst the others are optimistic. In the $1/-1$ setting(Figure 1b), the tweets with scores between -1 and 1 are not considered. Only those pictured with bright red and green are considered pessimistic and optimistic, respectively. The data pictured with pale red and green is ignored in the $1/-1$ setting.

Along these lines, it is easy to notice that even though the 0 settingexposes a somewhat larger dataset, the $1/-1$ setting may be solved easily due to the much radical expression of the data/labels.

1.3 HATE DATASET

One of the topics which is very active in the NLP research these days is Hate Speech Detection. Especially, there are multiple studies focusing on the presence of toxic language in tweets.

In our project we make use of the dataset proposed by Founta et al., 2018. They created a dataset of 80,000 tweets annotated as either: *normal*, *spam*, *abusive* or *hateful* (Figure 2).

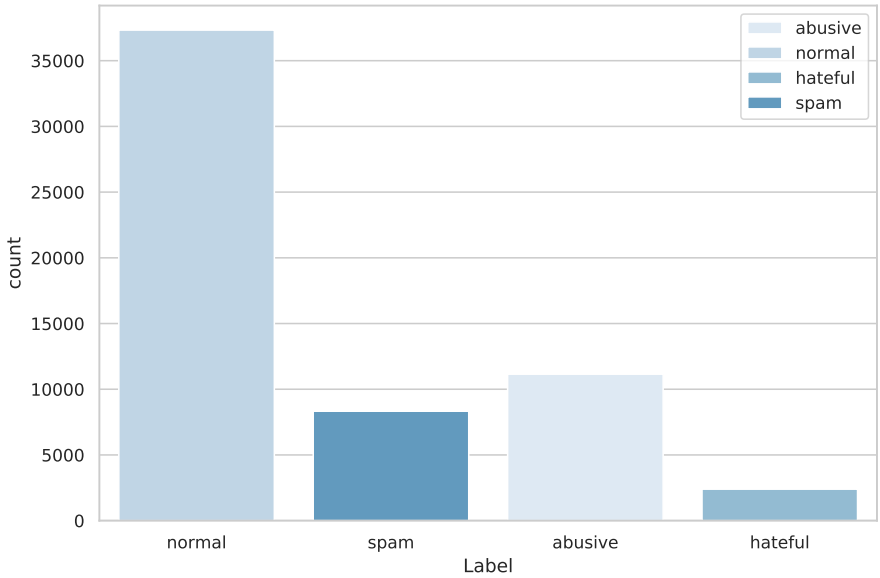


Figure 2: Label distribution in the Hate dataset, according to the found crawled tweets (replication of Figure 7 by Founta et al., 2018). Since most posted tweets are not offensive it is expected of datasets, targeting problems such as hate speech, to not be balanced. We can see that most tweets are labeled as normal.

The annotation procedure was an iterative one. First a broader list of labels was used in a crowdsourcing annotation phase. Second, by a statistical study, the most fuzzy labels were merged into more representative categories, such that the agreement would be a significant one.

1.4 EMONET DATASET

One example of dataset which approaches more thoroughly the granularity of sentiments was introduced by Abdul-Mageed and Ungar, 2017. In this work was analyzed a large spectrum of emotions, mainly those proposed in Plutchik, 1984. In Robert Plutchik's emotions taxon-

omy there eight primary emotions. These primary emotions are further structured into a total of 24 emotions coresponding to three levels of intensity of the primary ones, plutchik-1, plutchik-2 and plutchik-3.

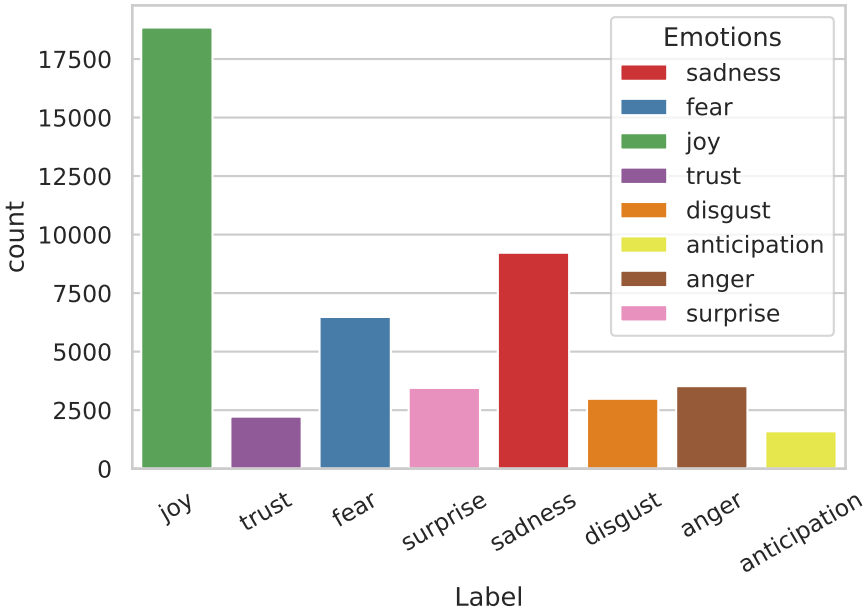


Figure 3: Label distribution in the EmoNet dataset by Abdul-Mageed and Ungar, 2017, according to the found crawled tweets. The labels correspond to the eight primary emotions. In particular, the eight emotions displayed above are sorted according to the intermediate circle of the Plutchik's wheel (PLUTCHIK-2).

The goal of the study proposed by Abdul-Mageed and Ungar, 2017 was to annotate a large dataset of tweets with Plutchik's emotions. The procedure/methodology followed for this annotation was one of distant supervision Mintz et al., 2009. On short, the procedure was as follows: A quarter billion tweets were crawled using *Tweeter's API*. The crawling procedure was performed by finding ones containing suggestive hashtags. Examples of such hashtags were synonyms of the emotions or related words. Afterwards, the irrelevant tweets (e.g. too short or non-English) were removed. The obtained dataset contained $1,608,233$ tweets.

Finally, a distant annotation methodology was pursued. The labeling was validated through a statistical study.

In our work we focused our attention on the available dataset constructed accordingly to plutchik-2 to which we refer as EmoNet. The emotions expressed/addressed by plutchik-2 are: *joy*, *trust*, *fear*, *surprise*, *sadness*, *disgust*, *anger*, and *anticipation* Plutchik, 1984.

This provided dataset $48,339$ contains distributed in the eight emotions as can be seen in Figure 3.

1.5 SENTIMENT DATASET

In the study presented by Caragea, Dinu, and Dumitru, 2018 there is an analysis of the correlation between optimism and sentiment polarity. As sentiment dataset they mainly use TSA[2].

The TSA dataset consists of $1,578,627$ tweets that are labeled as either negative or positive. Even though there is not a one-on-one correlation between sentiment polarity, optimism and pessimism, we shall see that it may be a strong correlation between them.

2 The TSA dataset can be found at the address: http://thinknook.com/twitter-sentiment-analysis-training-corpus-dataset-2012-09-21/

Part II

LINGUISTIC ANALYSIS OF OPTIMISM

Is somebody more devout to events long gone or is it fervent towards those to come? Does one tend to talk more about himself? Is the usage of hashtags bringing up to light insides on the writer?

In this chapter we will analyze multiple patterns in tweets and see to which degree are they related to the optimism or pessimism of the users. We shall see that indeed when using specific verb tense, polarity of words, special characters, etc. the optimism or pessimism of the user might be disclosed.

We shall explore statistical analysis on a number of linguistic features of tweets and analyze their correlation with optimism and pessimism.

2 LINGUISTIC FEATURES EXPRESSED IN OPTIMISM AND PESSIMISM

In the second part of this work we describe the seed of our project. Mainly, we summarize the work presented in Caragea, Dinu, and Dumitru, 2018. We will discuss both the results and our implementation with which we replicated the mentioned work.

The article has as main focus the analysis of optimism and pessimism in tweets using a deep learning methodology. The authors claim to that to their knowledge this was the first work in this direction.

The analysis is focused on the OPT dataset introduced in Ruan, Wilson, and Mihalcea, 2016. We discuss in this manuscript about the OPT dataset in Section 1.2.

The third part of the work presented in Caragea, Dinu, and Dumitru, 2018 was to analyze the linguistic structure of the tweets from the OPT dataset. We implemented the experiments proposed by them. Additionally, we further analyzed some additional linguistic features. Most of these features are insides that can be found in psychological studies.

As a summary, we investigated the correlation between optimism/pessimism and the following linguistically features:

1. Verb tenses, in Section 2.1;
2. Words polarity as negative or positive, in Section 2.2;
3. Parts of speech, in Section 2.3;
4. First person pronouns, in Section 2.4;
5. Special characters, in Section 2.5.

The first two (verb tense and words polarity) were proposed by Caragea, Dinu, and Dumitru, 2018. The following three were proposed by Caragea et al., 2020.

2.1 OPTIMISTS FOCUS ON THE FUTURE, PESSIMISTS ON THE PAST: VERB TENSE ANALYSIS

We thus first discuss, the correlation between verb tenses and optimism. Mainly, we investigated the frequencies of verb tenses in both optimistic and pessimistic tweets. We considered the past, present

and future verb tenses. In order to determine the tense of a verb we used a *parts of speech tagger* provided by the `spacy` library[1].

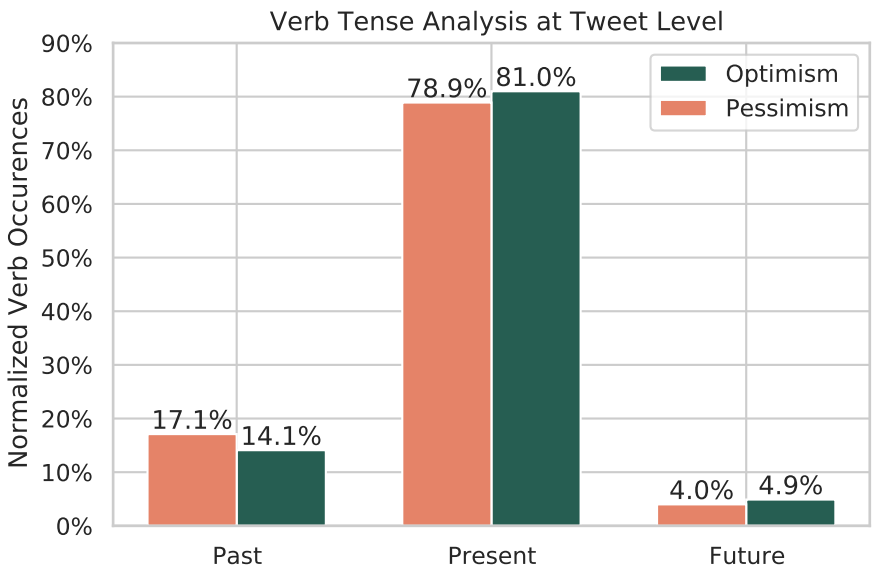


Figure 4: Correlation between optimism and verb tenses. The y-axis represents the percentage of tweets containing a specific verb tense. We can see that past tense is more present in pessimistic tweets. On the other hand, present and future tenses are more present in optimistic tweets.

As we can see in Figure 4, a tweet that is optimistic has a slightly higher chance to contain verbs in present and future form, compared to those who are pessimistic. Thus, 81% of the optimistic tweets contain verbs at present tense, compared to 78.9% of those who are pessimistic. With respect to verbs at future tense, they can be found in 4.9% of the optimistic tweets and only in 4% of the pessimistic tweets. On the reverse side, pessimistic tweets are prone to a more frequent usage of the verbs in past forms. Thus, 17.1% pessimistic tweets contain verbs on past tense, whilst only 14.1% of the optimistic ones do so.

The distribution of verb tenses on the pessimism and optimism is in accordance to the theory that pessimists tend to focus more on past events, whilst optimist people focus more on present and future events.

2.2 POLARITY WORDS USAGE

The second linguistic feature we analyzed was the presence of polarity words. Here we used the lexicons of negative and positive words, introduced by Hu and Liu, 2004, to see how many of them appear in optimistic and pessimistic tweets.

1 https://spacy.io/api/annotation#pos-tagging

In order to determine if a word is in a tweet we used a more general criterion than simple presence. Thus, for checking if a word w is contained in a tweet t we followed the following procedure:

1. We embedded the word w using GloVe pre-trained embeddings into the vector w_G;
2. We embedded the words of the tweet t using GloVe and sum their embeddings, obtaining the vector t_G;
3. If the cosine similarity between w_G and t_G was above 0.8, we considered that the word w is contained in the tweet t.

By using this procedure, additional information is captured. Thus, a word is contained in a tweet if its meaning/embedding is similar to the meaning/embedding of the tweet.

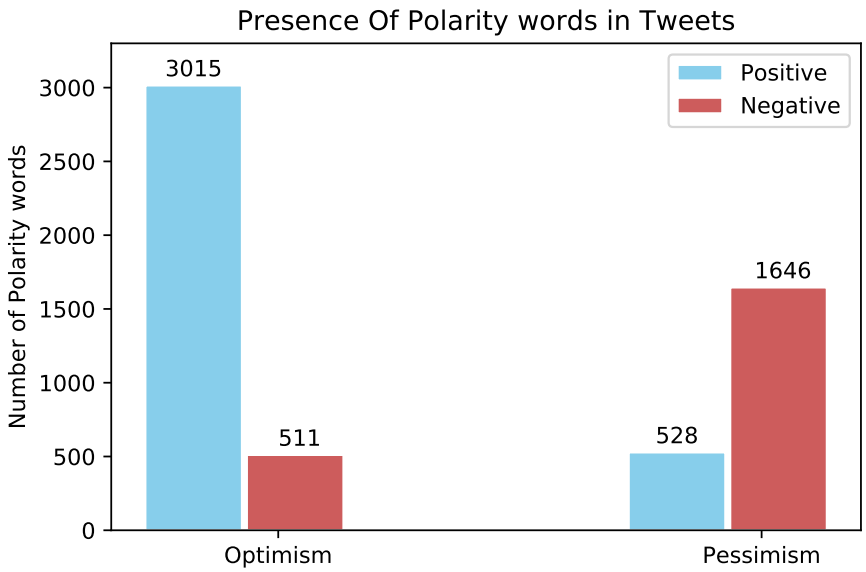


Figure 5: Correlation between optimism and polarity words frequency.

In Figure 5, we can see that optimistic tweets are characterized by having more positive words. Whilst pessimistic tweets tend to contain more negative words. Nevertheless, we can see the the proportion of positive words in pessimistic tweets is larger than the proportion of negative words in optimistic tweets. This fact suggests that there is not a one on one correlation between optimism and positivism and between pessimism and negativism. Concretely, even though a tweet has a good amount of positive words, it may still be a pessimistic tweet.

In Table 2, we can some of the most common polarity words in both optimistic and pessimistic tweets. We can see that optimistic tweets tend to contain more positive words. On the other hand pessimistic tweets have a much more smooth distribution of polarity words. Both

Optimism		Pessimism	
Positive	**Negative**	**Positive**	**Negative**
('love', 284)	('miss', 18)	('like', 145)	('f–k', 18)
('good', 215)	('funny', 17)	('right', 28)	('hate', 17)
('happy', 151)	('bad', 16)	('well', 22)	('shit', 16)
('like', 142)	('fall', 13)	('good', 21)	('fucking', 13)
('best', 100)	('hard', 10)	('work', 17)	('bad', 10)

Table 2: Most common polarity words used in optimistic/pessimistic tweets. Replication of Table 5 Caragea, Dinu, and Dumitru, 2018.

positive and negative words tend to appear at a similar frequency in pessimistic tweets.

2.3 PARTS OF SPEECH

We extended this analysis to more linguistic features.

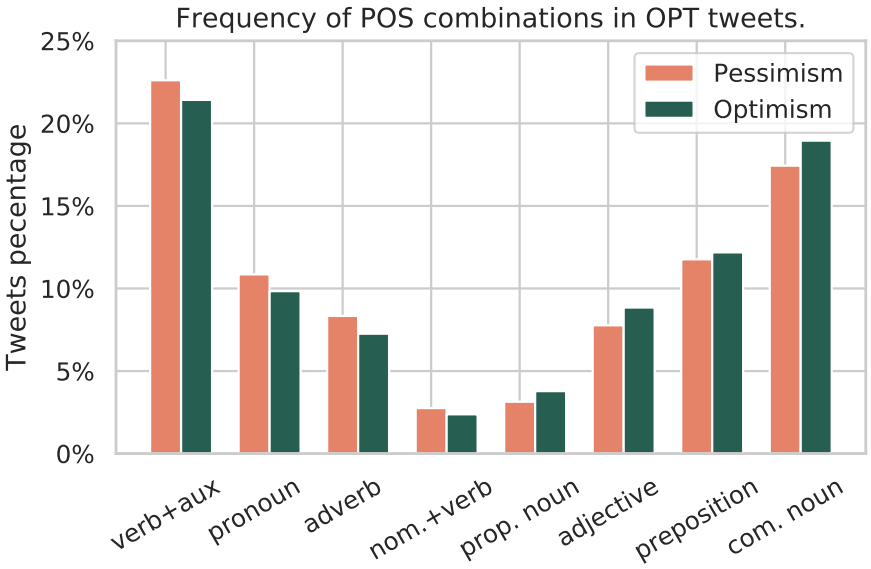
In this section we discuss the relationship between parts of speech and optimism. We used a parts of speech (POS) tagger specific for tweets (Owoputi et al., 2013).

In Figure 6, we can see that there are a number of parts of speech combinations that more specific than others for either optimism and pessimism. Thus, we notice the following combinations (also documented in Table 3) that are more specific to pessimistic tweets than for those who are optimistic:

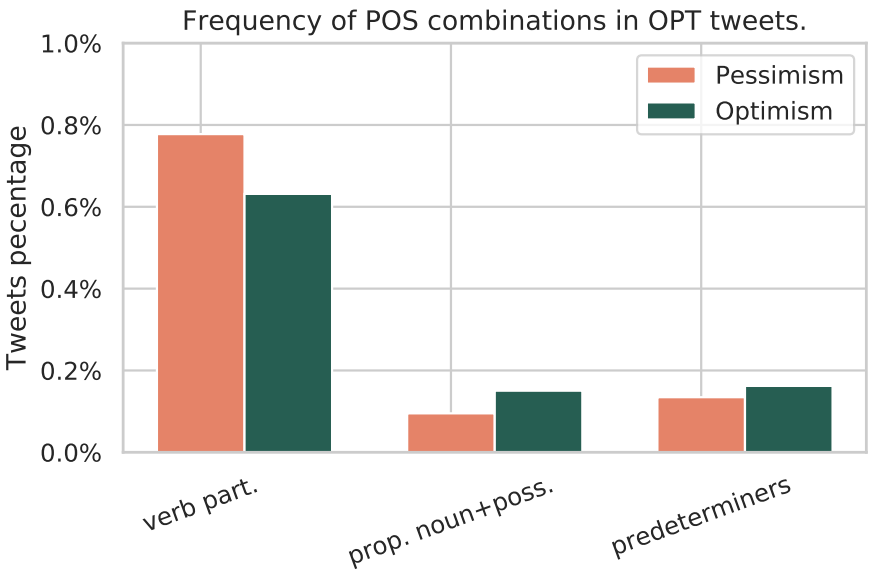
1. Verbs and auxiliaries;
2. Pronouns;
3. Adverbs;
4. Nominal verbs;
5. Verb particles.

We also notice the following parts of speech specific to optimistic tweets (see Table 3):

1. Proper Nouns, followed by a possessive;
2. Predeterminers;
3. Proper nouns;
4. Adjectives;
5. Prepositions;
6. Common nouns.



(a) Most common parts of speech.



(b) Less common parts of speech.

Figure 6: Correlation between parts of speech and optimism. In (a) we can see the most frequent parts of speech. In (b) There are displayed parts of speech that appear less than 1% in tweets.

Out of these results we can draw the inside that pessimistic tweets are more focused on actions. On the other hand, optimistic tweets tend to contain more descriptions (using more adjectives) and are more specific to objects and persons (using more nouns and pronouns).

Part of Speech	Pessimism Frequency	Optimism Frequency
	Pessimism Specific POS	
Verb and Auxiliary	**22.61**	21.42
Pronoun	**10.85**	9.84
Adverb	**8.34**	7.26
Nominal Verb	**2.76**	2.38
Verb Particle	**0.78**	0.63
	Optimism Specific POS	
Proper Noun + Possessive	0.10	**0.15**
Predeterminer	0.13	**0.16**
Proper Noun	3.14	**3.78**
Adjective	7.77	**8.85**
Preposition	11.77	**12.19**
Common Noun	17.44	**18.95**

Table 3: Correlation between optimism and parts of speech.

2.4 FIST PERSON PRONOUNS

Another inside drawn from the study proposed by Zimmermann et al., 2017 shows that depressive people tend to have more attention on themselves, emphasizing own fallibilities. Depressive attitude can be sometimes related also to pessimism. Thus, we were lead to investigate if references to the first person is an indicator of pessimism.

Indeed, as we can see in Figure 7, the frequency of first person pronouns in pessimistic tweets is of 41.92%, whilst in optimistic tweets this frequency is of 37.18%. Thus, pessimists use with over 11% more first person nouns.

This observation can be useful in the active research field of detecting depression. It is worthy to note that there might be some correlation between depression and pessimism.

2.5 SPECIAL CHARACTERS

There are some linguistic specificities of social media texts. One of them is the usage of some *special characters*, such as hashtags, emoticons or mentions.

In this section we analyze if there is a link between special characters and optimism and pessimism in tweets. Mainly, by special character we broadly understand non-textual fragments of tweets.

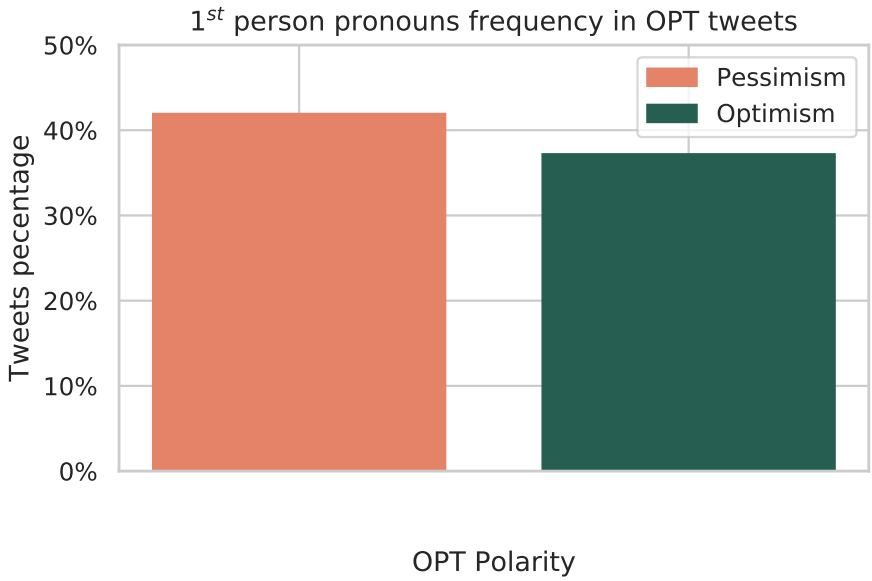


Figure 7: Correlation between first person pronouns and optimism.

As we can see in Figure 8 that there are indeed some correlations between the optimism of tweets and the special characters present in them. Thus, we observed that punctuation and other non-textual characters is more specific to pessimistic tweets. Whilst, for the optimistic ones there are more emoticons, hashtags and mentions. This reveals, among others, that fact that optimists are more likely to involve other members of the community in their conversations.

Special Character type	Pessimism Frequency	Optimism Frequency
Punctuation	**60.16**	56.85
Other	**4.43**	2.90
Emoticon	1.92	**3.35**
Hashtag	4.91	**7.47**
Mention	21.23	**22.49**

Table 4: Correlation between optimism and special characters.

The most noticeable features would be hashtags and emoticons. We can see in Table 4 that the number of optimistic tweets in which we can find emoticons is with 42.68% higher than in those that are optimistic. As well, hashtags appear in 34.32% more optimistic tweets than in those that are pessimistic.

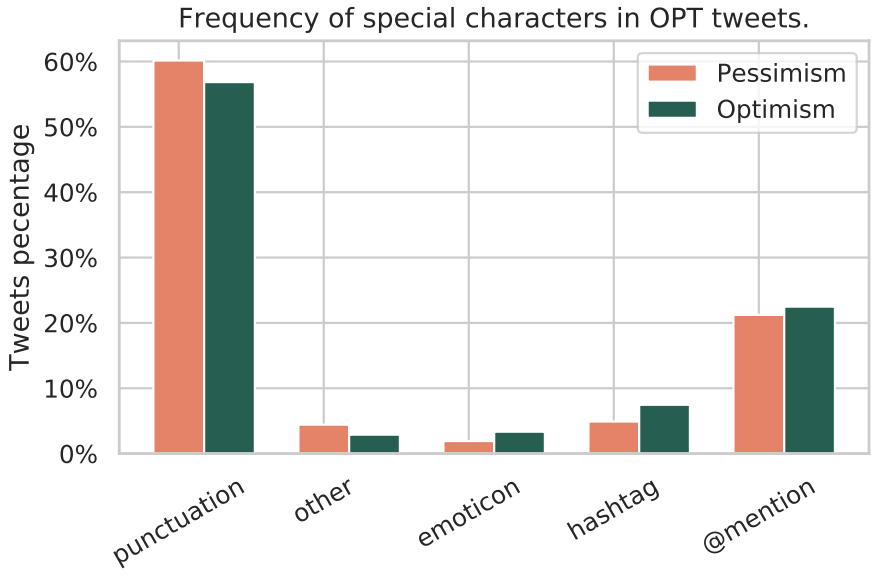


Figure 8: Correlations between special characters and optimism.

2.6 DISCUSSIONS

In this part we have seen how linguistic features may relate with optimism and pessimism.

We have seen that the dominant attributes, better characterizing pessimist are the usage of the past tense verbs, less positive words, more first person pronouns an punctuation. On the other side, optimists use more future tense verbs, positive words, hashtags and mentions.

While these features can give some intuition with respect to the inclination of tweet, there is no definitive single linguistic feature that we could expose in order to distinguish between optimism and pessimism.

Thus, in the following chapter, we will focus on building a deep learning model to solve the classification problem. By doing so, we will not focus on hand crafted features anymore.

Part III

DEEP LEARNING FOR OPTIMISM PREDICTION

While using deep learning for sentiment polarity prediction may seem as walking on a beaten track, there are adjacent narrow paths that are to least say unexplored. Concretely, many other feelings besides sentiment may be explored using deep learning methodologies.

In the third and final chapter of this manuscript, we analyze optimism and pessimism using a deep learning toolbox. We start by testing well established architectures, along pre-trained language models as we obtain a benchmark performance. We observe the high impact that the batch size may have on stochastic gradient descent learning, when the training dataset is small. In the end, we achieve a state of the art performance on the OPT dataset by using deep pre-trained attention based models.

We further explore the relationship between optimism, pessimism and other feelings. We thus transfer knowledge from datasets labeled on emotions, hate speech and sentiment polarity. Doing so, we further improve the performance of our models.

3

OPTIMISM AND PESSIMISM DISCRIMINATION USING DEEP LEARNING

In the third part of this manuscript we seek to draw a deep learning perspective over the detection of optimism and pessimism in *Twitter posts*.

We have seen in Chapter 2 that there are a couple of linguistic features that may hint towards either the optimism or pessimism of a tweet. Most of these features are found by following intuitions from psychological studies or from daily lives. Nevertheless, such peculiarities are prone to subjectivity. We know try to learn features by gradient descent, in order to be as objective as possible.

This chapter has the following outline[1]:

1. Deep Learning models to predict optimism: Section 3.1;
2. Train models on predicting sentiment polarity and test them on optimism prediction: Section 3.2;
3. Batch sizes influence: Section 3.3;
4. Attention based models for optimism prediction: Section 3.4;
5. Transfer learning from other feelings to optimism: Section 3.5;
6. Conclusions and future work Section 3.6.

3.1 OPTIMISM PREDICTION WITH DEEP LEARNING

We begin by addressing how can deep learning be used for optimism prediction. This problem was first analyzed by Caragea, Dinu, and Dumitru, 2018. They proposed a general framework for the model construction that can be summarized as follows:

1. Embed the tweets in a the vector space induced by the GloVe pretrained embeddings (Pennington, Socher, and Manning, 2014);
2. The embedded tweets are fed to a deep lerning model which has two main parts:
   a) The encoding part, consisting of various deep learning architectures;
   b) The prediction part, a three layer *multi layer perceptron*.

1 Points 1 and 2 where first proposed by Caragea, Dinu, and Dumitru, 2018. Points 3, 4 and 5 where also discussed by Caragea et al., 2020.

The used GloVe pre-trained embeddings were the version trained with 800 billion tokens. The usage of pre-trained language models, such as GloVe is of high importance since the OPT dataset is considerably small in order for a plain deep learning model to exhibit good performance.

The proposed encoders were *Bidirectional Long-Short Term Memory* (BiLSTM) Graves and Schmidhuber, 2005, one dimensional *Convolutional Neural Networks* (CNN) LeCun et al., 1989 and stack of *Gated Recurrent Units* (GRUStack) Cho et al., 2014. Besides these classical architectures, we also proposed the usage of a Transformer encoder (Vaswani et al., 2017).

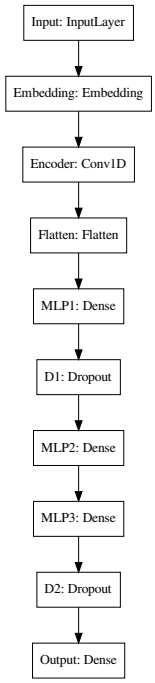


Figure 9: Basic deep learning model architecture for optimism prediction. The first two blocks make up the embedding part. The third and forth blocks represent the encoder part. The encoder part could consist of multiple architectures. The final blocks (fifth to tenth) represent the prediction part, a three layer multi layer perceptron.

We consider both settings of the optimism definition, discussed in the context of the OPT dataset (see Section 1.2). In the 0 setting, a tweet is considered optimistic if it has an average annotation grater than 0 and pessimistic otherwise. In the $1/-1$ setting, only tweets with average annotation greater than 1 (optimistic) and lower than -1 (pessimistic) are considered.

In Figure 10, we can see the substantial difference between the optimism definition settings. When the learning is performed in the context of the 0 setting we can see that the validation accuracy is un-

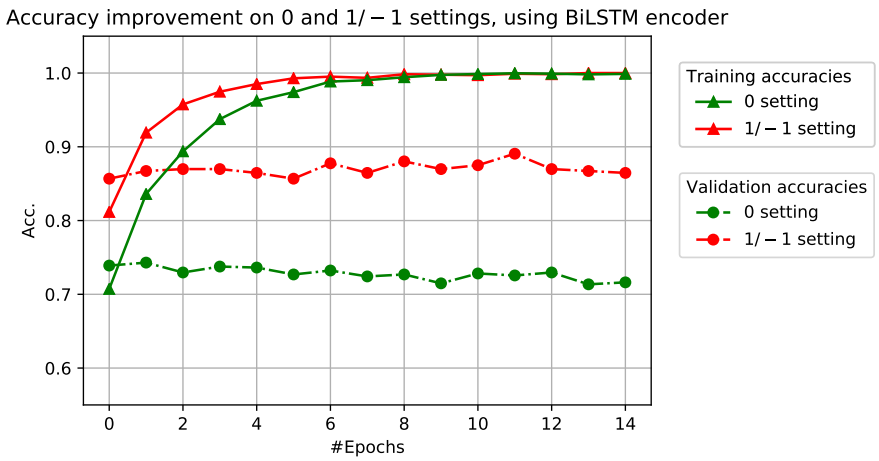


Figure 10: Comparison between 0 setting and $1/-1$ setting. The y-axis represents the accuracy of the model. The x-axis represents the number of training epochs passed. The depicted performance is of a BiLSTM encoder with a GloVe embedding. We can see that the difference between the two settings' accuracies is around 10%.

der 75%. In the context of the $1/-1$ setting the validation accuracy is over 85%. Another observation we can make is that most of the learning occurs during the first epoch. Since the OPT dataset is relatively small, one epoch seems to be enough for a model to learn most of the generalizing variation.

Model	Setting	
	0	$1/-1$
BiLSTM	74.56	85.42
CNN	74.70	84.90
GRUStack	76.04	85.94
Transformer	**78.45**	**90.36**
Naive Bayes	74.20	84.10
SVM	67.80	83.30

Table 5: Optimism prediction accuracy of deep learning models compared to standard machine learning models on both 0 setting and $1/-1$ setting. The highest accuracies are achieved by using a Transformer based model. Deep learning models exceed classical machine learning algorithms performances.

In Table 5 we present our results for all the models we have tested, compared to the reported classical machine learning models reported in Ruan, Wilson, and Mihalcea, 2016 and Caragea, Dinu, and Dumitru, 2018. We can see that the highest performance is achieved by using a Transformer encoding, on both the 0 setting and $1/-1$ setting.

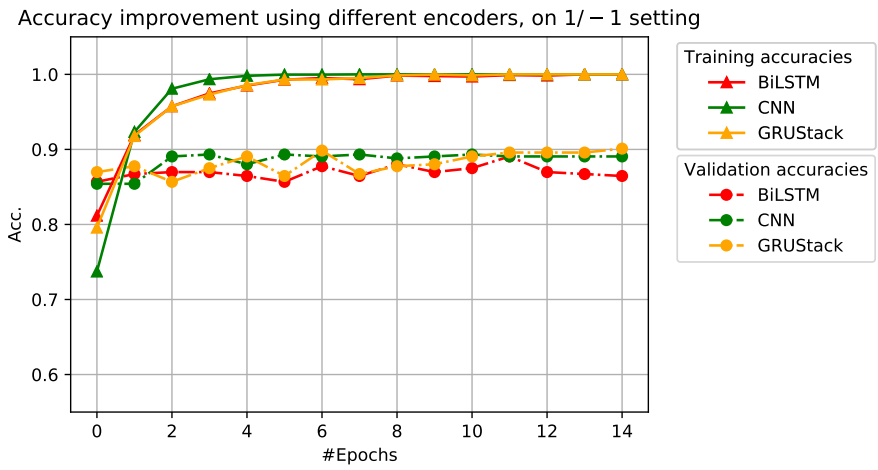


Figure 11: Comparison between BiLSTM, CNN and GRUStack encoder on the $1/-1$ setting. There is no substantial difference between the three types of encoders. All of them reach a validation accuracy close to 90%.

A comparison between the performance improvement of the encoders proposed by Caragea, Dinu, and Dumitru, 2018 is displayed in Figure 11 We can see that they have a very similar behavior. As well, the performance gap is within 2%.

We also experimented if other pre-trained embeddings would have an impact on the model performance. Thus, we also tested GloVe trained on tweeter data and FastText (Bojanowski et al., 2017).

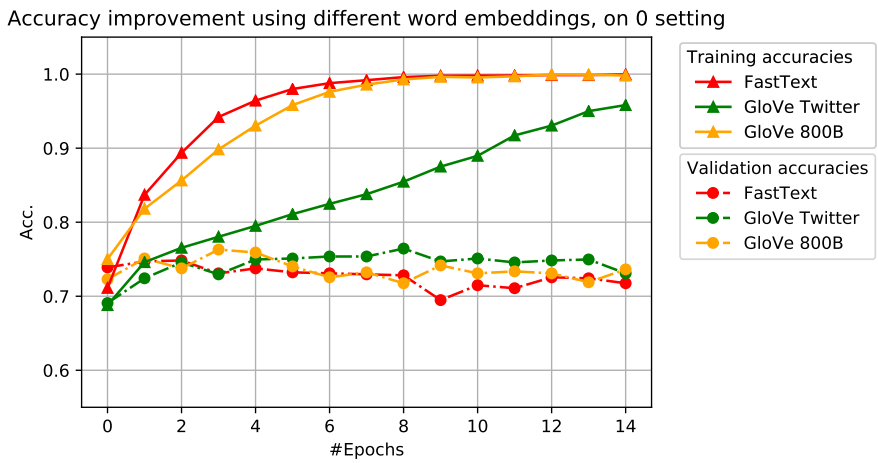


Figure 12: Comparison between pre-trained embeddings on the 0 setting when using a GRUStack encoder. The pre-trained embeddings are FastText, GloVe twitter and GloVe 800B. There is not a clear difference between using one of them. The validation accuracy of the models tend to be around 75%.

In Figure 12, we can see that the validation performance does not significantly change when using different pre-trained embeddings.

This fact suggests that contextual pre-trained embeddings such as GloVe and FastText could have some limitation for our problem. Thus, in Section 3.4 we will extend this analysis by using large scale contextual pre-trained embeddings, trying to improve our performance.

3.2 DISTINCTION BETWEEN SENTIMENT AND OPTIMISM

In this section we discuss the correlation between sentiment and optimism. This issue was also analyzed in Section 3.2 by Caragea, Dinu, and Dumitru, 2018. The idea was to train a model on a sentiment dataset and test it on the OPT dataset. As a sentiment dataset TSA (see Section 1.5) was used. The question is to what degree the polarity of a tweet is related to the optimism or pessimism in it.

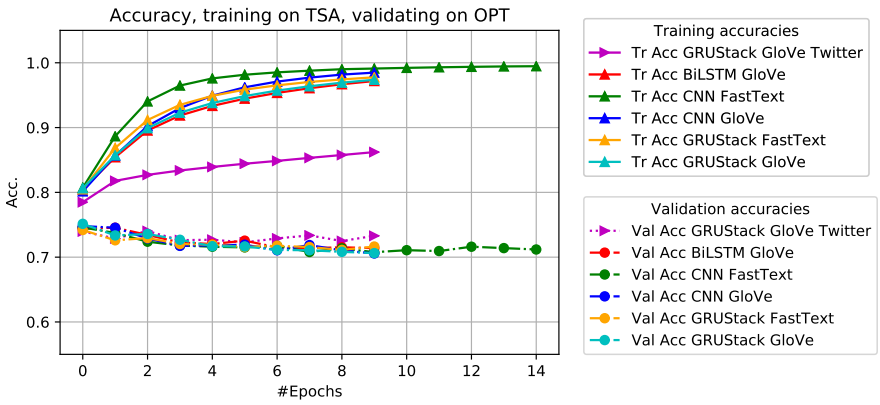


Figure 13: Optimism prediction improvement when training on the TSA dataset and testing on predicting optimism. The optimism setting considered is 0 setting. We can se that the meaningful training occurs in the first epoch. In the following epochs the models start to overfit on predicting sentiment polarity and decrease their accuracy on predicting optimism.

In Figure 13 we can see the performance improvement when using various models and pre-trained embeddings.

We displayed in Table 6 the accuracies of the models when trained either on OPT or TSA and tested on predicting optimism. We can see that even though a high accuracy is reached also when training on TSA, at least 4% of the variation in OPT is not explained by TSA. Still, the fact that we can obtained over 70% accuracy on OPT just by training on sentiment polarity prediction gives us some intuitions. First, there is an obvious correlation between specific feelings and optimism and pessimism. Second, we could try to extend the learning by either training on various related datasets and afterwards by fine-tuning on OPT. This ideas will be explored in Section 3.5.

Model	Accuracy	
	OPT	TSA
BiLSTM	74.56	71.40
CNN	74.70	70.57
GRUStack	**76.04**	70.62

Table 6: The accuracies of models when trained on OPT (left column) and on TSA (right column) and testing on predicting optimism. Whilst not seeing any tweets from the OPT dataset, we can see that training on predicting sentiment polarity reveals an accuracy of over 70%. This indicates a correlation between sentiment and optimism that may be explored.

3.3 IMPORTANCE OF BATCH SIZES VARIATION WHEN USING SMALL DATASETS

In this section we analyze the impact of the batch size in the performance improvement when learning on the OPT dataset. Since the OPT dataset is relatively small in size, having only $7,475$ samples in the 0 setting and $3,847$ in the $1/-1$ setting, there are some issues that may appear during the learning process.

When using stochastic gradient descent, the updates become more fuzzy as the used batch size increases. We observed a noticeable influence of the batch size in the performance when learning on OPT.

In Figure 14 we can see that as the batch size increases, the performance of the model decreases significantly, for the 0 setting. Thus, if for a batch size of 16, a model can reach an accuracy of almost 80%, when the batch size is $4,096$ the accuracy does not exceed 60%.

In Figure 15 we can see the impact of batch size variations in the context of the $1/-1$ setting. Here is even more visible that as the the batch size gets smaller, the accuracy is greater. Nevertheless, the advantage of a large batch size would be the smaller chance of overfitting the training set. Still, since the OPT dataset is small (it consists of less than two batches of $4,096$) there needs to be made the compromise of learning the variation in as few epochs as possible.

3.4 ATTENTION BASED MODELS TO ACHIEVE SOTA PERFORMANCE

After the proposal of the Transformer layer, by Vaswani et al., 2017, the language model performances based on attention started to exceed previous models on most of the natural language processing tasks. In this section we discuss our approach of using large, pre-trained, attention based models to predict optimism in tweets. The models

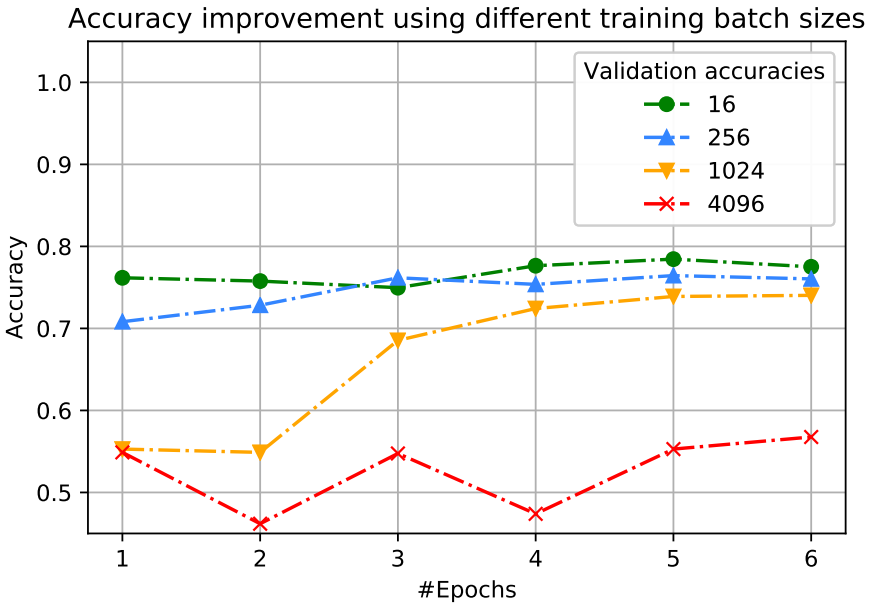


Figure 14: Impact of batch size variation when learning to predict optimism in 0 setting. Since the OPT dataset is small, a large batch size flattens the updates too much and hinders improvement. The displayed performance is of a Transformer encoder alongside GloVe pre-trained embeddings.

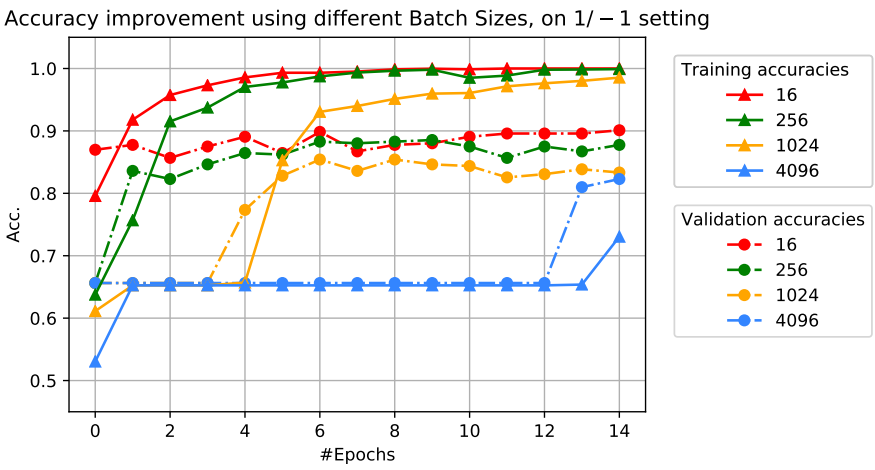


Figure 15: Impact of batch size variation when learning in 1/−1 setting. The validation accuracy increases as the batch size decreases. This effect is due to the small size of the OPT dataset. The displayed performance is of a BiLSTM encoder alongside GloVe pre-trained embeddings.

that we explored are ALBERT Lan et al., 2019, BERT Devlin et al., 2018, RoBERTa Liu et al., 2019 and XLNet Yang et al., 2019.

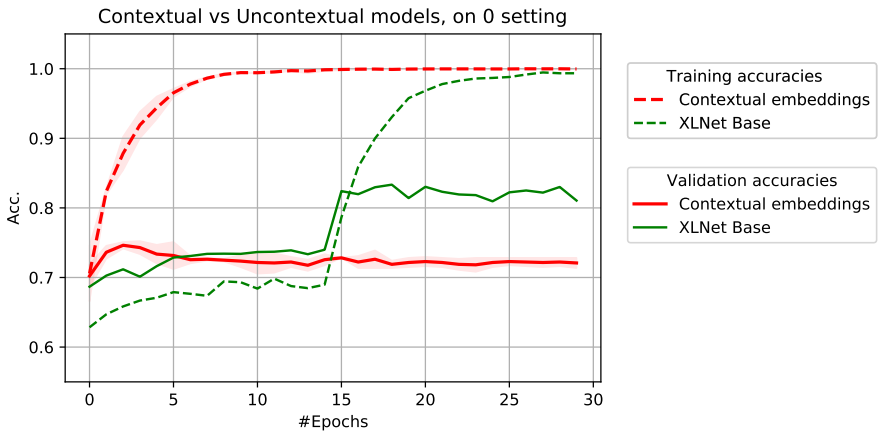


Figure 16: Improvement of XLNet compared to the mean of contextual models on predicting optimism when considering the 0 setting. Until epoch 15 the weights of XLNet are frozen. Afterwards, all the weights of XLNetare fine-tuned. Fine-tuning largely improves the model's performance.

Since the number of parameters of the attention based models we explored are much higher than previous ones[2], and the pre-training is richer and takes into account the context in which words appear, we expect that the performance of the attention based models to be higher.

In Figure 16 we can see the performance improvement of XLNet compared to the average of the previous models (BiLSTM, CNN and GRUStack), when predicting optimism, in 0 setting.

Model	Accuracies
GRUStack	80.19
ALBERT Base	81.71
BERT Base	81.40
BERT Large	81.79
RoBERTa Base	**83.17**
XLNet Base	81.71

Table 7: Performance achieved by attention based models, compared to the previous best model, GRUStack (as reported by Caragea, Dinu, and Dumitru, 2018), when predicting optimism in the 0 setting. The highest accuracy is achieved by RoBERTa base, with almost 3% over the previous state-of-the-art model.

The performance of all the attention based models we tested can be seen in Table 7. The sudden improvement occurring in epoch 15

2 For example the GRUStack model had under four million parameters, while the BERT model has over one hundred million parameters.

is caused by starting to fine-tuning the entire model. Until epoch 15, only weights of the prediction/final layer were set to be trainable, while the rest were frozen. This effect is more clearly displayed in Figure 18. The best performance was achieved by an ensemble model, in which we combined the prediction of all the five models and trained a linear classifier on top of them.

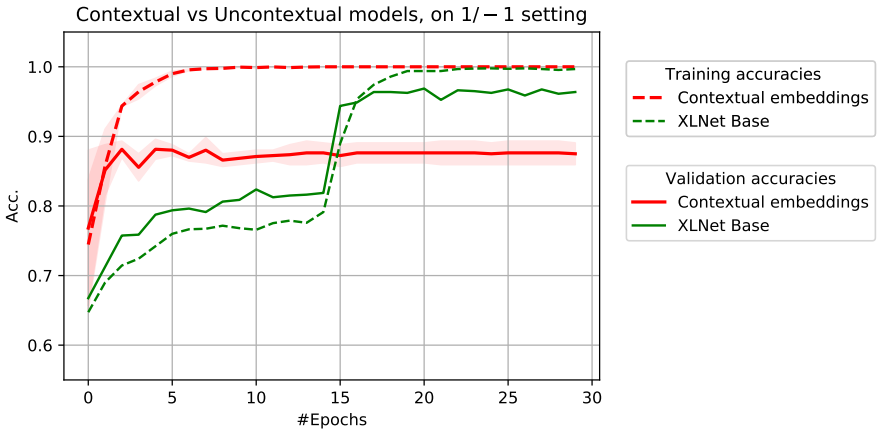


Figure 17: Optimism prediction performance improvement of XLNet compared to the mean of contextual models when considering $1/-1$ setting. Until epoch 15 only the top layer of XLNet was trainable. Afterwards, all the weights are set to be fine-tuned. This has a large impact on the final performance, increasing the accuracy with more than 10%.

In the case of the $1/-1$ settingthe improvement is even more visible, see Figure 17. Again, we can see the major improvement on the 15th epoch achieved by starting to fine-tune all the weights of the model, not just the final layer.

In Table 8 we can see the results of the attention based models, compared to the previous best reported one (CNN[3]), in the $1/-1$ setting. The improvement achieved by the RoBERTa is of almost 5%.

In Figure 18 we can see the significant effect of fine-tuning RoBERTaon the OPT dataset, in the 0 setting. The improvement brought by fine-tuning is of more than 10%. The same effect occurs for all the models, and in both settings. When keeping the weights frozen, the attention based models achieve an accuracy similar to that of the classical machine learning ones (Table 5). There are studies that specifically address when a deep learning model should be or not fine-tuned according to the details of the problem (Peters, Ruder, and Smith, 2019).

3.5 TRANSFER LEARNING FROM FEELINGS TO OPTIMISM

Previously, in Section 3.2, we have seen that there might been some variation in other datasets that could help generalize better on pre-

3 The best result reported by Caragea, Dinu, and Dumitru, 2018

Model	Accuracies	
	Validation	Test
CNN	90.32	
BERT Base	95.91	94.23
BERT Large	95.91	94.95
RoBERTa Base	95.19	**95.28**
XLNet Base	97.35	94.71
Best Ensemble	**97.84**	**95.67**

Table 8: Accuracies of attention base models compared with the previous best model (CNN) when predicting optimism on the $1/-1$ setting. The best performance is achieved by RoBERTa base with an improvement of almost 5% over the previous state-of-the-art model CNN (as reported by Caragea, Dinu, and Dumitru, 2018).

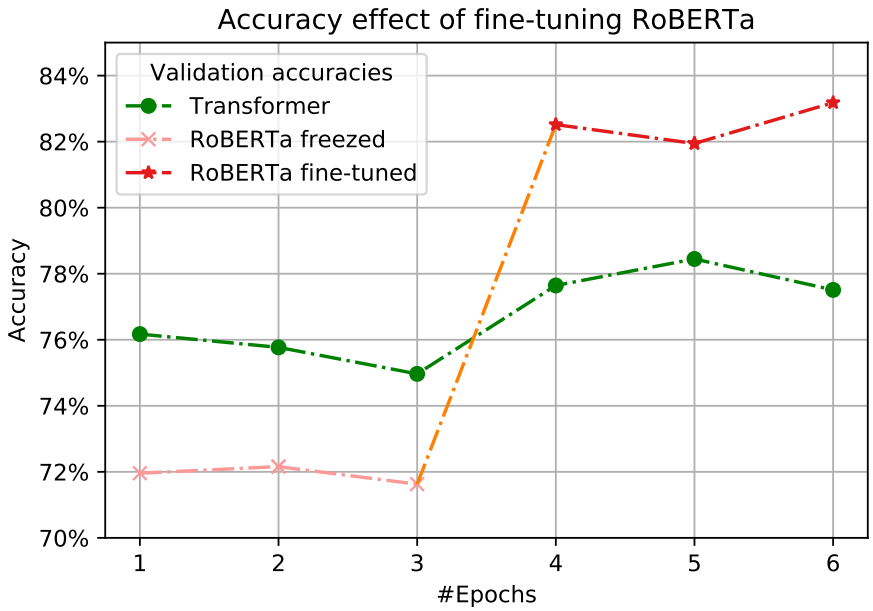


Figure 18: Improvement when fine-tuning compared with freezing big models. Until the third epoch only the top layer of RoBERTa is trainable. When fine-tuning the entire model, the accuracy increase is of more than 10%.

dicting optimism. The experiments already performed, trained models on the TSA dataset (a sentiment polarity dataset, see Section 1.5), and tested them on predicting optimism, in the OPT dataset, on the 0 setting.

A natural extension would be to start fine-tuning the models on the OPT dataset. As well, it is worth exploring further if there links between other feelings and optimism.

Here, besides the TSA dataset, we explored two additional dataset, EmoNet, a dataset with tweets labeled with eight primary emotions (see Section 1.4) and Hate, a dataset of tweets which are labeled as hateful or not (see Section 1.3).

Model	Pre-training dataset			
	Neither	EmoNet	Hate	TSA
ALBERT Base	81.71	80.49	81.45	81.69
BERT Base	81.40	80.85	81.95	81.56
BERT Large	81.79	81.25	81.58	81.74
RoBERTa Base	83.17	83.64	83.12	**83.67**
XLNet Base	81.71	81.32	82.29	81.19
Best Ensemble		**83.85**		

Table 9: Accuracies of attention based models after pre-traing them on other feelings datasets and fine tuning on OPT, on the 0 setting. The improvements are not substantial. Still, a new highest accuracy is achieved by pre-training RoBERTa base on predicting sentiment polarity and afterwards fine-tuning it on predicting optimsm.

In Table 9 we can see the results of transferring knowledge from the EmoNet, Hate and TSA datasets to optimism prediction, in the 0 setting, when using deep attention based models.

3.6 CONCLUSIONS AND FUTURE WORK

In this work we discussed multiple aspects of optimism and pessimism. There has not been an extensive focus on detecting optimism using computational linguistic tools. Out of the previous works exploring optimism and pessimism we can mention the proposal of the OPT dataset by Ruan, Wilson, and Mihalcea, 2016. As well, the study of Caragea, Dinu, and Dumitru, 2018 explores for the first time the application deep learning methodologies and gives some linguistic intuitions on the topic.

In this manuscript we have seen a survey of the previous approaches. In Chapter 1 we introduced the OPT dataset alongside other feelings datasets.

In Chapter 2 we have seen how can we characterize a tweet based on the linguistic features expressed in it. Thus, we have seen that there are a couple of peculiarities of both optimists and pessimists.

Finally, in Chapter 3 we presented a number of deep learning models that are able to learn to distinguish between optimistic and pessimistic tweets. By employing large pre-trained attention based models we achieved state of the art performance on the OPT dataset.

In future studies we believe that there are a number of directions that may be valuable to explore.

First, there is a need for building a larger dataset. To do so there are a number of studies describing distant supervision strategies (Abdul-Mageed and Ungar, 2017; Conforti et al., 2020; Felt and Riloff, 2020). Larger, qualitative, datasets may also increase the interest in studying this topic.

Second, the study of depression is a very important subject in the late years. We believe that there is room for analysis of correlation between optimism, pessimism and depression.

Third, we have seen that transfer learning is plausible for the problem of optimism detection. Nevertheless, we could improve the transfer strategy. It would be interesting to explore how does `GPT-3` (Brown et al., 2020) perform on this task, being a dedicated few-shot learning model.

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A

APPENDIX - IMPLEMENTATION DETAILS

The support code for our project is published and is publicly available on GitHub. It can be consulted at the following address:

`https://github.com/StefanCobeli/Exploring-Optimism`

The main libraries we used during the development process are:

- HuggingFace on top of Pytorch for the pre-trained attention based models (Section 3.4);
- Keras (Chollet et al., 2015) on top of Tensorflow (Abadi et al., 2016), for the classical deep learning models (Section 3.1);
- Pandas (McKinney, 2010), for the datasets' management;
- Seaborn (Waskom et al., 2017) and Matplotlib (Hunter, 2007) for the plots in this work;
- Spacy (Honnibal and Montani, 2017) for the linguistic analysis experiments (Chapter 2).

The LaTeXtemplate used for this manuscript was `classicthesis`, developed by André Miede. The template is available at:

`https://bitbucket.org/amiede/classicthesis/`

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